Agenda

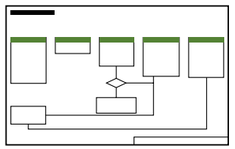
Open Orders by MSA report builder: team review, 1 hr 30 min.





Sheet 2

How it was made
VBA <=> Python



Sheet 3

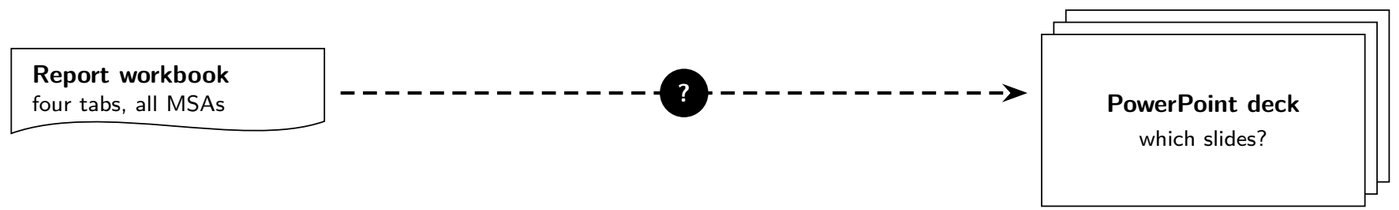
One run, step by step
The UNMAPPED loop

From Lexi's exports

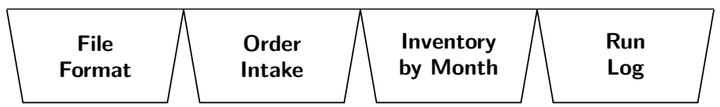
1. Exports into one folder
2. Check setup
3. Build report
4. Walk the four tabs

Questions as we go

Where does the data need to go?



What the build gives us: four tabs



File Format	Order Intake	Inventory by Month	Run Log
The rules, in writing	All MSAs on one tab	Units by MSA; pick the month	Every file read, with row counts

The data

- Is the Open Order export needed from snowflake?
- Where does the size class main file live?

The deck

- Which slides come from this report?
- What figures need to be included?
- Is the MSA analysis missing anything that needs to go into the deck?
- Editable charts on the template? One slide per MSA, or one summary?

We leave with: the slide list, each slide's required data, open questions assigned owners.

Title	Project	Macro	Rev.	Date	Drawn	Sheet
Agenda	**Open Orders by MSA**	**v2.8**	**13**	**2026-10-15**	**Gust F.**	**1 of 3**

Design Process

How the Open Orders by MSA report builder was made, in the order it was done.

Designed by hand. Repetitive code written with generative AI. Checked by two separate builds that must agree.

1 Map Inputs and Outputs

Done by hand

In

- Snowflake exports: Order Intake, Inventory by Month, Open Order
- The size class workbook

Out

- File Format, Order Intake, Inventory by Month, Run Log

Rules written down first

- What is filtered, hidden, sorted and counted
- Those rules became the File Format tab

2 Analyze the Input Files

Done by hand, with Copilot

- 10 to 15 short prompts on each file
- Read the headers, MSA spellings, order statuses, inventory months and model groups
- Each prompt: a fixed reply format, most with a row-count check
- What the real exports look like, checked against what the macro expects

3 Design Before Code

Done by hand

Architecture

- Two builds of the same rules: VBA for the team, Python as the reference to check it against

Macro

- Load, check setup, apply rules, build
- The setup check stops the run before anything is created
- Config sheet: nine settings, five buttons; nobody edits code

Test harness

- A data contract every report is checked against
- Report checker, VBA import check, comparison with the hand-built tab

Test data, built to break it

- Traps from the real exports and known Excel pitfalls: CANCELLED spelled two ways, `" mx "` with spaces, dates stored as text, an unknown model, an old report and a lock file in the folder
- Expected answers checked against hand-computed results

4 Fill In the Boilerplate

Generative AI, reviewed by hand

- The repetitive code inside each part: file reading, formatting, print setup, logging
- Every result checked against the rules and the tests

5 Prove It Out

VBA macro
`build-open-orders-report.bas`
2,836 lines

=

Python reference
`build_report_reference.py`
940 lines

Same inputs and the same rules, written separately in two languages. Both build identical reports: every value, hidden row, evaluated summary formula and Run Log line (`check_report.py -compare`, 19 checks, on three builds).

- **Test data:** every expected answer hit (22 and 22 checks); Austin sample: 15 of 15 checks
- **Hand-built Austin tab:** 170 of 170 order rows and 29 of 29 inventory rows, in the same size-class order (small TOTAL differences from the anonymised sample)
- **Found by the tests and fixed:** an `IIf` that ran both branches; state left over from an earlier run; a blank list separator that stripped the commas from a formula

Next

- First build in the team's own Excel (25-minute test protocol)
- The team's own size class list replaces the starter list
- The report feeds the PowerPoint deck

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Design Process	**Open Orders by MSA**	**v2.8**	**13**	**2026-10-15**	**Gust F.**	**2 of 3**

MSA Open Orders Reporting Macro – User Manual

The macro does these for every MSA (no longer done by hand):

- Filter to CTL and MX
- Hide cancelled orders behind a filter
- Assign a size class per model
- Sort by size class, with a line between classes
- Show one inventory month (change it in the drop-down)
- Total the CTL and MX units in stock per MSA

Before you start

- First time only: import the module into a saved .xlsm workbook and enable macros
- Save Snowflake exports as .xlsx (.xlsm and .xls also work; CSV files are not read)
- Set the size class file on Config (Choose size class file): if it is blank, every build stops

1 Load Snowflake Exports

In one folder

`Order Intake*.xlsx` (required)

`Inventory by Month*.xlsx` (optional)

`Open Order*.xlsx` (listed, not used) ?

Open question: is this export needed? Order Intake already has an `IS_OPEN_ORDER` *column.*

2 Open Config Sheet

- First time: press Alt+F8, run `ShowOpenOrdersReportConfig`
- Choose input folder, output folder (blank: the input folder) and size class file
- Click Build report

3 Check Setup

- Runs on every Build report
- Reads every file
- Checks the settings, both folders, the master list, each export's columns
- Lists the MSAs and months found

Problems?

No → 4 Apply Build Rules

Yes →

Not built

- One list of what to fix, in a message and in the Last run box
- No report is created

4 Apply Build Rules

Same rules for every MSA

- Every file of a kind merged
- MSA from the MSA column, else the file name
- Keep CTL, MX, SSL
- Size class from the master list only; a model not on it is **UNMAPPED**
- Sort: size class, order status, newest date, MSA
- Cancelled and SSL rows kept, hidden by filters

5 Create Workbook

Four tabs

- File Format (the rules, and this build)
- Order Intake (all MSAs)
- Inventory by Month (units by MSA for the month in the drop-down; every month's detail)
- Run Log

Saved to the output folder. If the file is there, Excel asks to overwrite; No saves a time-stamped copy.

Last run box on Config: Report built, or Report built – *n* items to look at (warnings, e.g. an export skipped for a missing column; the report is still built).

Size class workbook

- The team's master list: model group to size class
- A new model or size class is a row here, not a code change

If **UNMAPPED**, add the model to the list, save it and build again.

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